



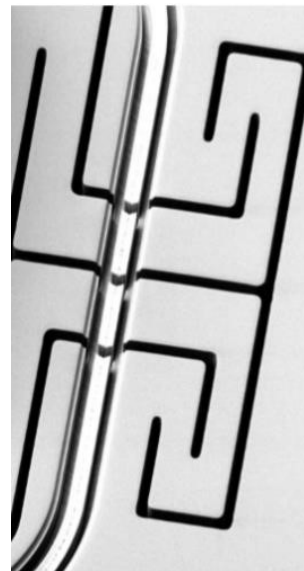
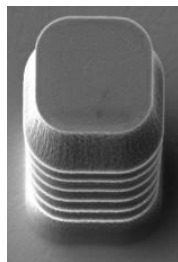
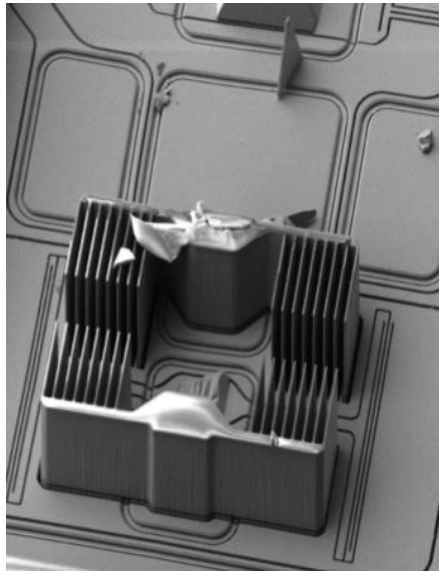
一起感知^{无线}美好世界
Sensing the world together

QMA6100P水平校准说明文档

上海矽睿科技有限公司
QST CORPORATION

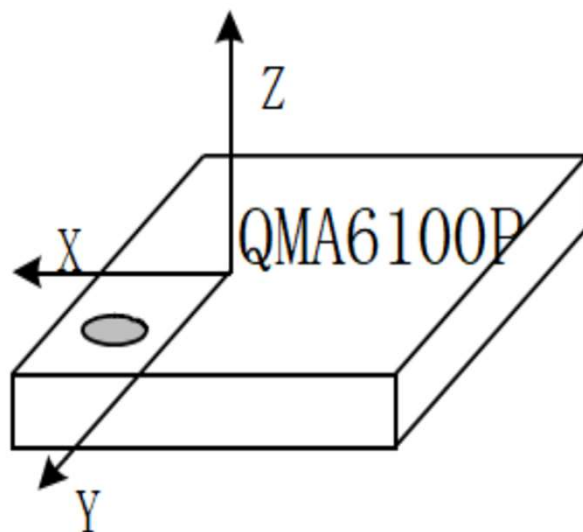


前言



上海矽睿科技有限公司
QST CORPORATION

QMA6100P零偏offset校准方法



- 1、将带有QMA6100P的模组放到水平面上，确保模组水平且无振动；
- 2、执行校准算法`QstAccel_perform_calibration(float acc[3])`，读取QMA6100P原始加速度计数据，参考下文代码；
- 3、校准完成后，保存加速度零偏offset值`offset_acc`；
- 4、校准后的加速度计值为 $\text{acc} = \text{acc_raw} - \text{offset_acc}$ 。

QMA6100P零偏offset校准函数

```
void QstAccel_perform_calibration(float acc[3])
{
    unsigned char    buf_reg[6];
    short            raw_acc_xyz[3];
    float            acc_raw[3];
    uint8_t axis = 0;
    union byte_float
    {
        float float_data;
        uint8_t byte[4];
    } float_to_byte;

    ret = qma6100p_readreg(QMA6100P_XOUTL, buf_reg, 6);
    if(ret == QMA6100P_FAIL)
    {
        QMA6100P_ERR("read xyz read reg error!!!\n");
        return QMA6100P_FAIL;
    }
    raw_acc_xyz[0] = (short)(((buf_reg[1]<<8)|(buf_reg[0])));
    raw_acc_xyz[1] = (short)(((buf_reg[3]<<8)|(buf_reg[2])));
    raw_acc_xyz[2] = (short)(((buf_reg[5]<<8)|(buf_reg[4])));

    raw_acc_xyz[0] = raw_acc_xyz[0]>>2;
    raw_acc_xyz[1] = raw_acc_xyz[1]>>2;
    raw_acc_xyz[2] = raw_acc_xyz[2]>>2;

    //m/s2
    acc_raw[0] = (float)(raw_acc_xyz[0] * ONE_G) / acc_lsb_div;
    acc_raw[1] = (float)(raw_acc_xyz[1] * ONE_G) / acc_lsb_div;
    acc_raw[2] = (float)(raw_acc_xyz[2] * ONE_G) / acc_lsb_div;

    if(imu_init != 1)//flash中已经存储校准标记?
    {
        if(cali_count == 0)
        {
            memset((void *)accel_calibration_sum, 0, sizeof(accel_calibration_sum));
            cali_count++;
        }
        else if(cali_count < MAX_CALI_COUNT)
```

QMA6100P零偏offset校准函数

```
for(axis = 0; axis < 3; axis++)
{
    if(axis == 2)
    {
        accel_calibration_sum[axis] += (acc_raw[axis] - ONE_G);
    }
    else
    {
        accel_calibration_sum[axis] += acc_raw[axis];
    }
}
cali_count++;
}
else if(cali_count == MAX_CALI_COUNT)
{
    for(axis = 0; axis < 3; axis++)
    {
        offset_acc[axis] = (accel_calibration_sum[axis] / (MAX_CALI_COUNT - 1));
    }
    accel_init = 1;

    float_to_byte.float_data = offset_acc[0];
    Calibration_data[0] = float_to_byte.byte[0];
    Calibration_data[1] = float_to_byte.byte[1];
    Calibration_data[2] = float_to_byte.byte[2];
    Calibration_data[3] = float_to_byte.byte[3];

    float_to_byte.float_data = offset_acc[1];
    Calibration_data[4] = float_to_byte.byte[0];
    Calibration_data[5] = float_to_byte.byte[1];
    Calibration_data[6] = float_to_byte.byte[2];
    Calibration_data[7] = float_to_byte.byte[3];

    float_to_byte.float_data = offset_acc[2];
    Calibration_data[8] = float_to_byte.byte[0];
    Calibration_data[9] = float_to_byte.byte[1];
    Calibration_data[10] = float_to_byte.byte[2];
    Calibration_data[11] = float_to_byte.byte[3];

    StoreCalibrationToFile();

    cali_count++;
}
}
```

QMA6100P零偏offset校准函数

```
else          //有存储，直接计算
{
    for(axis = 0; axis < 3; axis++)
    {
        acc[axis] = acc_raw[axis] - offset_acc[axis];
    }
}
```

THANK YOU



上海总部

地址：上海市浦东新区海科路99号6号楼3层
邮编：201210
电话：+86 21 6951 7300
网址：www.qstcorp.com
商务洽谈：business@qstcorp.com

研发中心

硅谷：120 Baytech Drive, San Jose, California 95134 USA

办事处

深圳：广东省深圳市宝安区海秀路23号龙光世纪大厦A座
1101室
香港：香港黄竹坑道42号利美中心20楼9室 电话：+852
2648-0386/+852 2648-0389